AA256 HW1 Solution

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1 Problem 1

The general expression for determining the stress transformation matrix for a given rotation is as follows:

$$[\sigma'] = [\beta][\sigma][\beta]^{\mathrm{T}} \quad (1)$$

where [β] is defined as the fundamental rotation matrix. Note that the expression for the stress transformation matrix can be found in Appendix 1 of the course reader.

1.1 Rotation about the x-axis

For a rotation about the x-axis, the fundamental rotation matrix $[\beta]_x$ is shown below:

$$[\beta]_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & cos\,\theta & sin\,\theta \\ 0 & -sin\,\theta & cos\,\theta \end{bmatrix}$$

From equation 1, we can determine the stress transformation matrix for a rotation about the x-axis.

$$[\sigma'] = [\beta]_x[\sigma][\beta]_x^{\mathrm{T}} \quad (2)$$

Plugging in $[\beta]_x$ to equation 2, the expanded version of the equation can be seen:

$$\begin{bmatrix} \sigma'_{11} & \sigma'_{12} & \sigma'_{13} \\ \sigma'_{21} & \sigma'_{22} & \sigma'_{23} \\ \sigma'_{31} & \sigma'_{32} & \sigma'_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & cos\,\theta & sin\,\theta \\ 0 & -sin\,\theta & cos\,\theta \end{bmatrix} \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & cos\,\theta & -sin\,\theta \\ 0 & sin\,\theta & cos\,\theta \end{bmatrix}$$

After the matrix multiplication is completed, the following expressions for each of the rotated stress components can be shown:

$$\sigma'_{11} = \sigma_{11}$$

$$\sigma'_{22} = \sigma_{22}\cos^2\theta + 2\sigma_{23}\cos\theta\sin\theta + \sigma_{33}\sin^2\theta$$
$$\sigma'_{33} = \sigma_{33}\cos^2\theta - 2\sigma_{23}\cos\theta\sin\theta + \sigma_{22}\sin^2\theta$$
$$\sigma'_{23} = \sigma_{23}(\cos^2\theta - \sin^2\theta) + \cos\theta\sin\theta(\sigma_{33} - \sigma_{22})$$
$$\sigma'_{13} = \sigma_{13}\cos\theta - \sigma_{12}\sin\theta$$
$$\sigma'_{12} = \sigma_{12}\cos\theta + \sigma_{13}\sin\theta$$

Combining like terms from Voight notation and creating a matrix formulation of the system from the stress vectors, the solution results:

$$\begin{Bmatrix}\sigma'_1\\ \sigma'_2\\ \sigma'_3\\ \sigma'_4\\ \sigma'_5\\ \sigma'_6\end{Bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0 & 0 & 0\\ 0 & \cos^2\theta & \sin^2\theta & 2\cos\theta\sin\theta & 0 & 0\\ 0 & \sin^2\theta & \cos^2\theta & -2\cos\theta\sin\theta & 0 & 0\\ 0 & -\cos\theta\sin\theta & \cos\theta\sin\theta & \cos^2\theta-\sin^2\theta & 0 & 0\\ 0 & 0 & 0 & 0 & \cos\theta & -\sin\theta\\ 0 & 0 & 0 & 0 & \sin\theta & \cos\theta\end{bmatrix}\begin{Bmatrix}\sigma_1\\ \sigma_2\\ \sigma_3\\ \sigma_4\\ \sigma_5\\ \sigma_6\end{Bmatrix}$$

From this expression, $[T]_x$ is:

$$[T]_x = \begin{bmatrix}1 & 0 & 0 & 0 & 0 & 0\\ 0 & \cos^2\theta & \sin^2\theta & 2\cos\theta\sin\theta & 0 & 0\\ 0 & \sin^2\theta & \cos^2\theta & -2\cos\theta\sin\theta & 0 & 0\\ 0 & -\cos\theta\sin\theta & \cos\theta\sin\theta & \cos^2\theta-\sin^2\theta & 0 & 0\\ 0 & 0 & 0 & 0 & \cos\theta & -\sin\theta\\ 0 & 0 & 0 & 0 & \sin\theta & \cos\theta\end{bmatrix} \quad (3)$$

1.2 Rotation about the y-axis

For a rotation about the y-axis, the fundamental rotation matrix $[\beta]_\gamma$ is shown below:

$$[\beta]_\gamma = \begin{bmatrix}\cos\theta & 0 & -\sin\theta\\ 0 & 1 & 0\\ \sin\theta & 0 & \cos\theta\end{bmatrix}$$

From equation 1, we can determine the stress transformation matrix for a rotation about the y-axis.

$$[\sigma'] = [\beta]_\gamma[\sigma][\beta]_\gamma^T \quad (4)$$

Plugging in $[\beta]_\gamma$ to equation 4, the expanded version of the equation can be seen:

$$\begin{bmatrix} \sigma'_{11} & \sigma'_{12} & \sigma'_{13} \\ \sigma'_{21} & \sigma'_{22} & \sigma'_{23} \\ \sigma'_{31} & \sigma'_{32} & \sigma'_{33} \end{bmatrix} = \begin{bmatrix} cos\,\theta & 0 & -sin\,\theta \\ 0 & 1 & 0 \\ sin\,\theta & 0 & cos\,\theta \end{bmatrix} \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix} \begin{bmatrix} cos\,\theta & 0 & sin\,\theta \\ 0 & 1 & 0 \\ -sin\,\theta & 0 & cos\,\theta \end{bmatrix}$$

After the matrix multiplication is completed, the following expressions for each of the rotated stress components can be shown:

$$\sigma'_{11} = \sigma_{11}\,cos^2\,\theta - 2\sigma_{13}\,cos\,\theta\,sin\,\theta + \sigma_{33}\,sin^2\,\theta$$

$$\sigma'_{22} = \sigma_{22}$$

$$\sigma'_{33} = \sigma_{33}\,cos^2\,\theta + 2\sigma_{13}\,sin\,\theta\,cos\,\theta + \sigma_{11}\,sin^2\,\theta$$

$$\sigma'_{23} = \sigma_{23}\,cos\,\theta + \sigma_{21}\,sin\,\theta$$

$$\sigma'_{13} = \sigma_{13}(cos^2\,\theta - sin^2\,\theta) + \sigma_{11}\,sin\,\theta\,cos\,\theta - \sigma_{33}\,sin\,\theta\,cos\,\theta$$

$$\sigma'_{12} = \sigma_{12}\,cos\,\theta - \sigma_{23}\,sin\,\theta$$

Combining like terms from Voight notation and creating a matrix formulation of the system from the stress vectors, the solution results:

$$\begin{Bmatrix} \sigma'_1 \\ \sigma'_2 \\ \sigma'_3 \\ \sigma'_4 \\ \sigma'_5 \\ \sigma'_6 \end{Bmatrix} = \begin{bmatrix} cos^2\,\theta & 0 & sin^2\,\theta & 0 & -2cos\,\theta\,sin\,\theta & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ sin^2\,\theta & 0 & cos^2\,\theta & 0 & 2cos\,\theta\,sin\,\theta & 0 \\ 0 & 0 & 0 & cos\,\theta & 0 & sin\,\theta \\ cos\,\theta\,sin\,\theta & 0 & -cos\,\theta\,sin\,\theta & 0 & cos^2\,\theta - sin^2\,\theta & 0 \\ 0 & 0 & 0 & -sin\,\theta & 0 & cos\,\theta \end{bmatrix} \begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \sigma_4 \\ \sigma_5 \\ \sigma_6 \end{Bmatrix}$$

From this expression, $[T]_\gamma$ is:

$$[T]_\gamma = \begin{bmatrix} cos^2\,\theta & 0 & sin^2\,\theta & 0 & -2cos\,\theta\,sin\,\theta & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ sin^2\,\theta & 0 & cos^2\,\theta & 0 & 2cos\,\theta\,sin\,\theta & 0 \\ 0 & 0 & 0 & cos\,\theta & 0 & sin\,\theta \\ cos\,\theta\,sin\,\theta & 0 & -cos\,\theta\,sin\,\theta & 0 & cos^2\,\theta - sin^2\,\theta & 0 \\ 0 & 0 & 0 & -sin\,\theta & 0 & cos\,\theta \end{bmatrix} \quad (5)$$

2 Problem 2

2.1 Stress transformation

The rotations act on the coordinate axes, successively about X_1, X_2', and X_3''. The passive component transformation for each step uses the rotation matrices from Problem 1 and the course reader. Thus the combined direction-cosine matrix is:

$$[\beta] = [\beta z(45°)][\beta_Y(30°)][\beta_x(90°)] \quad (6)$$

The contracted ordering is (11, 22, 33, 23, 13, 12), so the initial stress vector is:

$$\{\sigma\} = \{1, 3, 6, 5, 5, 2\}^T \, psi$$

For a symmetric tensor A, each step uses $A' = \beta A \beta^T$. Constructing the corresponding 6×6 matrices and composing them in the same order gives:

$$[T^\sigma] = [Tz(45°)][T_Y(30°)][T_x(90°)]$$

$$[T^\sigma] = \begin{bmatrix} 0.3750 & 0.1250 & 0.5000 & 0.5000 & 0.8660 & 0.4330 \\ 0.3750 & 0.1250 & 0.5000 & -0.5000 & -0.8660 & 0.4330 \\ 0.2500 & 0.7500 & 0 & 0 & 0 & -0.8660 \\ -0.3062 & 0.3062 & 0 & -0.6124 & 0.3536 & 0.3536 \\ 0.3062 & -0.3062 & 0 & -0.6124 & 0.3536 & -0.3536 \\ -0.3750 & -0.1250 & 0.5000 & 0 & 0 & -0.4330 \end{bmatrix} \quad (7)$$

$$\{\sigma'''\} = [T^\sigma]\{\sigma\} = \begin{Bmatrix} 11.4462 \\ -2.2141 \\ 0.7679 \\ 0.0254 \\ -2.6136 \\ 1.3840 \end{Bmatrix} psi \quad (8)$$

Values are rounded to four decimal places after computing the combined transformation at full precision.

2.2 Engineering strain transformation

The given off-diagonal tensor entries must be doubled to form the engineering strain vector. The numerical tensor is unchanged, but its contracted engineering vector is:

$$\{\varepsilon\} = \{1, 3, 6, 10, 10, 4\}^{\mathrm{T}}$$

Let the subscript t identify the contracted tensor-strain vector, with tensor shear entries. The Reuter matrix converts it to engineering strain:

$$\{\varepsilon\} = [R]\{\varepsilon_t\}, \quad [R] = diag(1, 1, 1, 2, 2, 2)$$

$$[T^{\varepsilon}] = [R][T^{\sigma}][R]^{-1} \quad (9)$$

$$[T^{\varepsilon}] = \begin{bmatrix} 0.3750 & 0.1250 & 0.5000 & 0.2500 & 0.4330 & 0.2165 \\ 0.3750 & 0.1250 & 0.5000 & -0.2500 & -0.4330 & 0.2165 \\ 0.2500 & 0.7500 & 0 & 0 & 0 & -0.4330 \\ -0.6124 & 0.6124 & 0 & -0.6124 & 0.3536 & 0.3536 \\ 0.6124 & -0.6124 & 0 & -0.6124 & 0.3536 & -0.3536 \\ -0.7500 & -0.2500 & 1.0000 & 0 & 0 & -0.4330 \end{bmatrix}$$

$$\{\varepsilon'''\} = [T^{\varepsilon}]\{\varepsilon\} = \begin{Bmatrix} 11.4462 \\ -2.2141 \\ 0.7679 \\ 0.0508 \\ -5.2271 \\ 2.7679 \end{Bmatrix}$$

3 Problem 3

3.1 Plane Stress

For the plane stress condition, it is known that σ_{33}, σ_{23}, and σ_{13} are equal to zero because there are no stresses in the out of plane direction.

Starting with the compliance matrix for isotropic materials, we can apply the assumptions for the plane stress conditions.

$$\begin{bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \epsilon_{33} \\ 2\epsilon_{23} \\ 2\epsilon_{13} \\ 2\epsilon_{12} \end{bmatrix} = \frac{1}{E}\begin{bmatrix} 1 & -\nu & -\nu & 0 & 0 & 0 \\ -\nu & 1 & -\nu & 0 & 0 & 0 \\ -\nu & -\nu & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2(1+\nu) & 0 & 0 \\ 0 & 0 & 0 & 0 & 2(1+\nu) & 0 \\ 0 & 0 & 0 & 0 & 0 & 2(1+\nu) \end{bmatrix}\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{23} \\ \sigma_{13} \\ \sigma_{12} \end{bmatrix}$$

$$\epsilon_{11} = \frac{1}{E}(\sigma_{11} - \nu\sigma_{22})$$

$$\epsilon_{22} = \frac{1}{E}(-\nu\sigma_{11} + \sigma_{22})$$

$$\epsilon_{33} = \frac{1}{E}(-\nu\sigma_{11} - \nu\sigma_{22})$$

$$2\epsilon_{12} = \frac{2(1+\nu)}{E}\sigma_{12}$$

Put this back into a 3x3 matrix with only in plane terms. Note that we do not include the ϵ_{33} expression because it is fully dependent on the in-plane terms.

$$\begin{bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ 2\epsilon_{12} \end{bmatrix} = \frac{1}{E}\begin{bmatrix} 1 & -\nu & 0 \\ -\nu & 1 & 0 \\ 0 & 0 & 2(1+\nu) \end{bmatrix}\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{bmatrix}$$

To find the stiffness matrix, we take the inverse of the compliance matrix.

$$\begin{bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{12}\end{bmatrix} = \begin{bmatrix}\frac{E}{1-\nu^2} & \frac{\nu E}{1-\nu^2} & 0\\ \frac{\nu E}{1-\nu^2} & \frac{E}{1-\nu^2} & 0\\ 0 & 0 & \frac{E}{2(1+\nu)}\end{bmatrix}\begin{bmatrix}\epsilon_{11}\\ \epsilon_{22}\\ 2\epsilon_{12}\end{bmatrix}$$

Simplify this expression to the following:

$$\begin{bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{12}\end{bmatrix} = \frac{E}{1-\nu^2}\begin{bmatrix}1 & \nu & 0\\ \nu & 1 & 0\\ 0 & 0 & \frac{1-\nu}{2}\end{bmatrix}\begin{bmatrix}\epsilon_{11}\\ \epsilon_{22}\\ 2\epsilon_{12}\end{bmatrix}$$

The stiffness matrix for the plane stress condition for steel is then:

$$[C]plane\ stress\ = \frac{E}{1-\nu^2}\begin{bmatrix}1 & \nu & 0\\ \nu & 1 & 0\\ 0 & 0 & \frac{1-\nu}{2}\end{bmatrix} \quad (10)$$

3.2 Plane Strain

For the plane strain condition, it is known that ϵ_{33}, ϵ_{23}, and ϵ_{13} are equal to zero because there are no strains in the out of plane direction.

Starting with the stiffness matrix for isotropic materials, we can apply the assumptions for the plane strain conditions.

$$\begin{bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{33}\\ \sigma_{23}\\ \sigma_{13}\\ \sigma_{12}\end{bmatrix} = \begin{bmatrix}\lambda+2\mu & \lambda & \lambda & 0 & 0 & 0\\ \lambda & \lambda+2\mu & \lambda & 0 & 0 & 0\\ \lambda & \lambda & \lambda+2\mu & 0 & 0 & 0\\ 0 & 0 & 0 & \mu & 0 & 0\\ 0 & 0 & 0 & 0 & \mu & 0\\ 0 & 0 & 0 & 0 & 0 & \mu\end{bmatrix}\begin{bmatrix}\epsilon_{11}\\ \epsilon_{22}\\ \epsilon_{33}\\ 2\epsilon_{23}\\ 2\epsilon_{13}\\ 2\epsilon_{12}\end{bmatrix}$$

where λ and μ are given as:

$$\lambda\ =\ \frac{E\nu}{(1+\nu)(1-2\nu)},\quad \mu\ =\ G\ =\ \frac{E}{2(1+\nu)}$$

$$\sigma_{11}\ =\ (\lambda+2\mu)\epsilon_{11}\ +\ \lambda\epsilon_{22}$$

$$\sigma_{22}\ =\ \lambda\epsilon_{11}\ +\ (\lambda+2\mu)\epsilon_{22}$$

$$\sigma_{33}\ =\ \lambda\epsilon_{11}\ +\ \lambda\epsilon_{22}$$

$$\sigma_{12}\ =\ 2\mu\epsilon_{12}$$

Put this back into a 3x3 matrix with only in plane terms. Note that we do not include the σ_{33} expression because it is fully dependent on the in-plane terms.

$$\begin{bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{12}\end{bmatrix} = \begin{bmatrix}\lambda+2\mu & \lambda & 0\\ \lambda & \lambda+2\mu & 0\\ 0 & 0 & \mu\end{bmatrix}\begin{bmatrix}\epsilon_{11}\\ \epsilon_{22}\\ 2\epsilon_{12}\end{bmatrix}$$

Substituting the equations for λ and μ:

$$\begin{bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{12}\end{bmatrix} = \begin{bmatrix}\frac{E(1-\nu)}{(1+\nu)(1-2\nu)} & \frac{E\nu}{(1+\nu)(1-2\nu)} & 0\\ \frac{E\nu}{(1+\nu)(1-2\nu)} & \frac{E(1-\nu)}{(1+\nu)(1-2\nu)} & 0\\ 0 & 0 & \frac{E}{2(1+\nu)}\end{bmatrix}\begin{bmatrix}\epsilon_{11}\\ \epsilon_{22}\\ 2\epsilon_{12}\end{bmatrix}$$

We can simplify to the final form of the stiffness matrix expression for stress and strain in isotropic materials:

$$\begin{bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{12}\end{bmatrix} = \frac{E}{(1+\nu)(1-2\nu)}\begin{bmatrix}1-\nu & \nu & 0\\ \nu & 1-\nu & 0\\ 0 & 0 & \frac{1-2\nu}{2}\end{bmatrix}\begin{bmatrix}\epsilon_{11}\\ \epsilon_{22}\\ 2\epsilon_{12}\end{bmatrix}$$

We can show that the stiffness matrix for the plane strain assumption in isotropic materials is as follows:

$$[C]plane\ strain = \frac{E}{(1+\nu)(1-2\nu)}\begin{bmatrix}1-\nu & \nu & 0\\ \nu & 1-\nu & 0\\ 0 & 0 & \frac{1-2\nu}{2}\end{bmatrix} \quad (11)$$